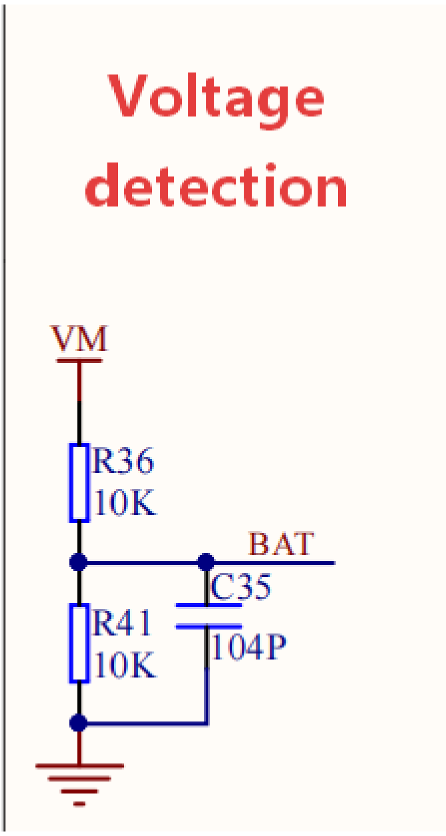
Voltage Collection

Experimental Purpose

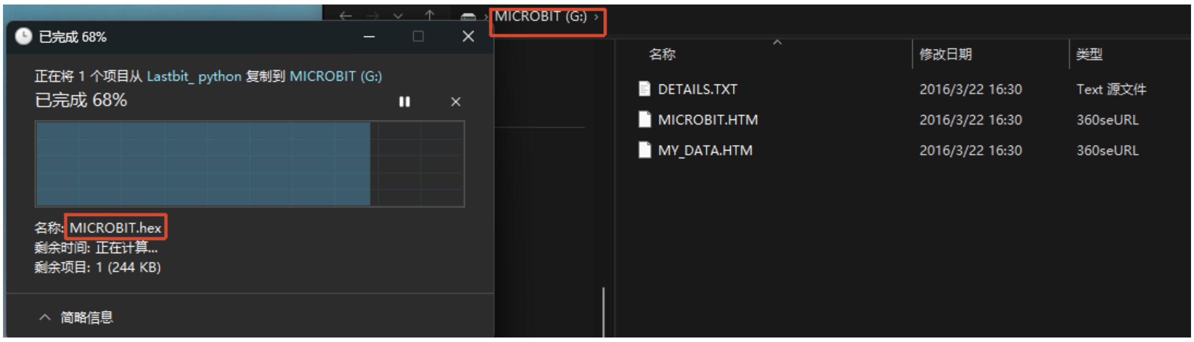
The Lastbit car is equipped with a single 18650 battery. In this section, we will learn how to read the battery voltage.

The expansion board uses a voltage-dividing voltage detection circuit. The parallel filter capacitors can absorb spike noise, providing a relatively stable and clean DC voltage for subsequent voltage sampling, and improving the anti-interference capability of the measurement.



Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**

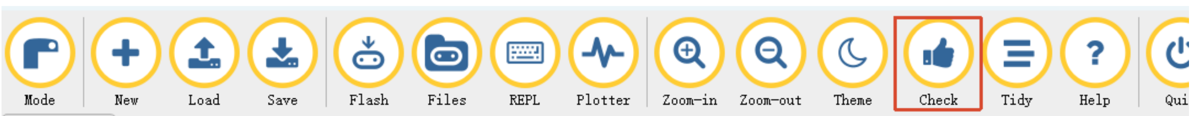
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.

   Click the `New` button in the toolbar at the top left, and a file titled untitled will appear.

3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

Whether or not you save does not affect the code check and flashing.

4. At this point, you can click `Check` to check whether the code format has any problems.



5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show `Copied code onto micro:bit.` at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.

6. Unplug the Microusb data cable and switch the switch to the ON position.

Code Implementation

```
# 4.Voltage_Collection

from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

while True:

    # Read battery voltage
    battery_voltage = Lastbit.read_battery()

    if battery_voltage is not None:

        # Show battery voltage
        display.scroll(str(battery_voltage) + "V")

    sleep(100)
```

`read_battery()`: reads the returned voltage value, which is a floating-point number. The unit is volts (V).

Experimental Phenomenon

The Microbit LED matrix will continuously scroll to display the read voltage value.